Team Control Number

15504

Problem Chosen

A

2024

HiMCM

Summary Sheet

The Olympic Games, a quadrennial global sporting spectacle, continually evolves with certain sports/disciplines/events (SDEs) being added or removed. The International Olympic Committee (IOC) employs six criteria to evaluate potential additions. The problem requires us to develop a model based on factors in line with IOC's criteria, predicting the inclusion of SDEs in future Olympics.

The core challenge of this problem is a binary classification task: determining whether a specific SDE should be included or excluded from the Olympics. As constrained by a limited number of SDEs per Olympic Games, the selection process prioritizes SDEs based on their relative significance. Probability offers a robust tool to assess the likelihood of an SDE's inclusion in alignment with Olympic values.

We developed a logistic regression model that incorporates three key items (Problem 2): a base probability (P_0) that evaluates the adherence to IOC's values based on nine quantitative and qualitative subcriteria we developed, an inertia component (P_{in}) accounting for the effect of past inclusion of an SDE in the Olympics, and a bias-correction item (P_{it}) that offers the flexibility to adjust the model best match the historical data. We developed nine subcriteria and collected historical data on Olympics sports from the Internet and literatures (Problem 1). Our model demonstrates strong predictive accuracy, accurately reproducing historical inclusion patterns, such as continuous presence, newly added or removed (Problem 3).

Sensitivity analysis revealed that the Popularity and Relevance and Innovation criteria are most influential factors in determining an SDE's likelihood of inclusion (Problem 5). According to our model prediction, in the 2032 Brisbane Olympics, Australian football takes the most potential to be included, followed by Zwift cycling and netball, among a total of 7 candidate sports including most popular ones in Australia (Problem 4). As Zwift virtual cycling becomes more popular among young people, it is more likely to be included in 2036 or beyond (Problem 4).

Compared to traditional methods such as the Analytical Hierarchy Process, our approach offers a more objective and flexible framework. Unlike subjective expert judgements, our model learns and constrains the weights and parameters from historical data to identify patterns and trends. While data collection remains a significant challenge, future efforts in collecting more comprehensive data can further enhance the model's predictive power. Our current model operates at the discipline level, it can be extended to the event level provided event-specific data is available. This would allow for even more granular predictions and insights into the future of the Olympic Games.

Table of Contents

1. Introduction

1.1 Background

The Olympic Games, a global sporting spectacle, have evolved significantly since their inception Athens, Greece in 1896. The International Olympic Committee (IOC) continuously adapts the Sports, Disciplines, or Events (SDEs) to reflect societal changes and maintain the Game's relevance. Over the past century, the Olympic program has expanded dramatically, incorporating a diverse range of sports (Figure 1), such as the recent addition of skateboarding, sport climbing, and surfing in the 2020 Tokyo Olympics. Conversely, some SDEs, like rope climbing, have been dropped due to factors such as declining popularity or high costs. Interestingly, some sports have been revived, such as baseball and softball, which will make a comeback at the 2028 Los Angeles Olympics after a 20-year absence.

To guide the selection of SDEs, the IOC employs six criteria: Popularity and Accessibility, Gender Equity, Sustainability, Inclusivity, Relevance and Innovation, and Safety and Fair Play. With these criteria provide a qualitative framework, a quantitative approach can future facilitate the decision-making process.

By framing the SDE selection problem as a binary classification task, we can leverage probabilistic methods to prioritize SDEs based on their relative importance. This approach allows us to model the complex interplay of factors that influence an SDE's inclusion in the Olympic Games, such as historical trends, global popularity, and alignment with Olympic values. Here, our team aims to refine the IOC's criteria and develop a model that can accurately predict the inclusion of SDEs in future Olympic Games.

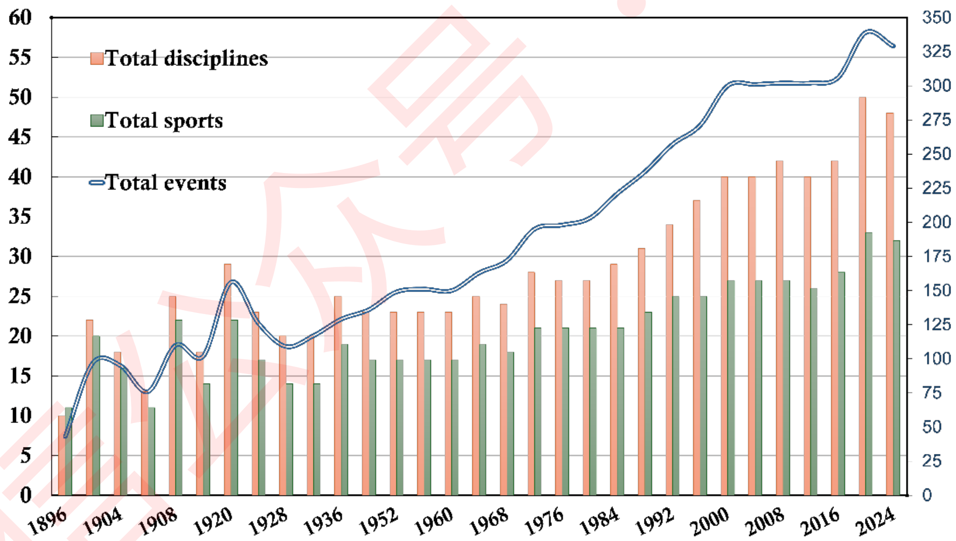

Figure 1 Evolution of Sports/Disciplines/Events (SDEs) in 1896-2024 Olympic Games (data source: HiMCM_Olympic_Data.xlsx)

1.2 Problem Restatement

Problem 1: Identifying key factors for SDE evaluation

What are the specific, measurable factors that influence the IOC's decision-making process for including or excluding SDEs in the Olympic Games? These factors should be categorized as quantitative or qualitative, constant or variable, and deterministic or probabilistic. Justify the selection of these factors and provide relevant units of measurement where applicable.

Problem 2: Developing a predictive model

Create a mathematical model that can accurately predict the likelihood of an SDE being included in future Olympic Games. The model should incorporate the identified factors and align with the IOC's established criteria.

Problem 3: Model validation

Evaluate the model's performance by testing it on a diverse set of SDEs. This should include both recently added and removed SDEs (e.g., from the 2020, 2024, and 2028 Olympics) and long-standing SDEs (e.g., those included since 1988 or earlier). Analyze the model's ability to accurately predict the current status of these SDEs.

Problem 4: Predicting future SDEs

Identify three SDEs that are potential candidates for inclusion in the 2032 Brisbane Olympics. Rank these SDEs based on their predicted likelihood of inclusion. Additionally, estimate on potential SDEs for future Olympic Games in 2036 or beyond.

Problem 5: Sensitivity analysis

Conduct a sensitivity analysis to assess the impact of various factors on the model's predictions. Identify the most influential factors and discuss the potential implications for the model's robustness and reliability.

Problem 6: Communicating findings to the IOC

Prepare a concise and informative one-page report summarizing the key findings and recommendations. Clearly explain the model's methodology, its strengths and limitations, and the specific SDEs that are predicted to be strong candidates for future inclusion or exclusion.

1.3 Problem Analysis

For Problem 1: While the IOC outlines six primary criteria, these are often qualitative in nature and difficult to directly quantify. To address this challenge, we will identify specific, measurable subcriteria, either quantitative or qualitative, that align with the IOC's broader objectives.

Data availability is a significant constraint. While the contest itself did not provide specific data, we will need to gather relevant information from various sources, such as historical Olympic data, sports statistics, and media reports. Given the challenges in obtaining comprehensive data for both sports and events, we will focus our analysis on disciplines. This approach allows us to balance the need for a sufficient sample size with the complexity of modeling the inclusion of SDEs.

For Problem 2: Traditional methods like the Analytical Hierarchy Process (AHP) or Analytic Network Process (ANP) could be used to evaluate SDEs based on the established criteria. However, these methods rely heavily on expert judgment and lack the ability to learn from historical data. This can limit their predictive accuracy.

The inclusion or exclusion of an SDE in the Olympics is essentially a binary classification problem. As constrained by a limited number of SDEs per Olympic Games, the selection process prioritizes SDEs based on their relative importance. By applying probabilistic methods, we can assess the likelihood of an SDE's inclusion, considering factors aligning with IOC's values. Logistic regression is a suitable statistical method to model this binary classification problem, using the identified factors as input variables

However, we must also consider the temporal dependency of SDE inclusion, as we can notice from the provided file HiMCM_Olympic_Data.xlsx. Historical data suggests that there is an "inertia effect," where the inclusion of an SDE in previous Olympics increases its chances of future inclusion. This effect is not accounted for in logistical regression models. We will adapt the model to account for this effect.

While training the logistic regression model on historical data significantly improved its overall accuracy, it is important to address the inherent constraints of the Olympic Games. For example, we will limit the total number of SDEs that can be included in a given Olympics. By incorporating these constraints, we can improve the model's accuracy and relevance. In other words, by enforcing constraints to the model, we have the ability to best match the patterns of SDE inclusion.

For Problems 3 and 4: Our collected data are absent for years after 2016. We have to assume that the historical trends observed from 1900 to 2016 are indicative of future trends. To validate the model and make predictions for the 2020, 2024, and 2028 Olympics, as well as future Games, we will employ appropriate extrapolation techniques.

For Problem 5: Sensitivity analysis will be conducted to assess the impact of different factors on the model's predictions. We will examine the sensitivity of the model to changes in the input factors. Additionally, we will investigate the sensitivity of the model to changes in its parameters.

1.4 Our work

Our workflow is illustrated in Figure 2. First, we identify the IOC’s six criteria by developing specific subcriteria and collecting relevant data from public sources (Problem 1). Next, we develop a predictive model that incorporates logistic regression, inertia, and bias correction to assess the likelihood of an SDE’s inclusion (Problem 2). We then validate the model’s performance by testing it on historical data and predicting the inclusion of SDEs in various inclusion/exclusion patterns for recent and future Olympics (Problems 3 and 4). We perform a sensitivity analysis to identify the most influential factors (Problem 5), and finally, we submit a report to the IOC summarizing our findings and recommendations (Problem 6).

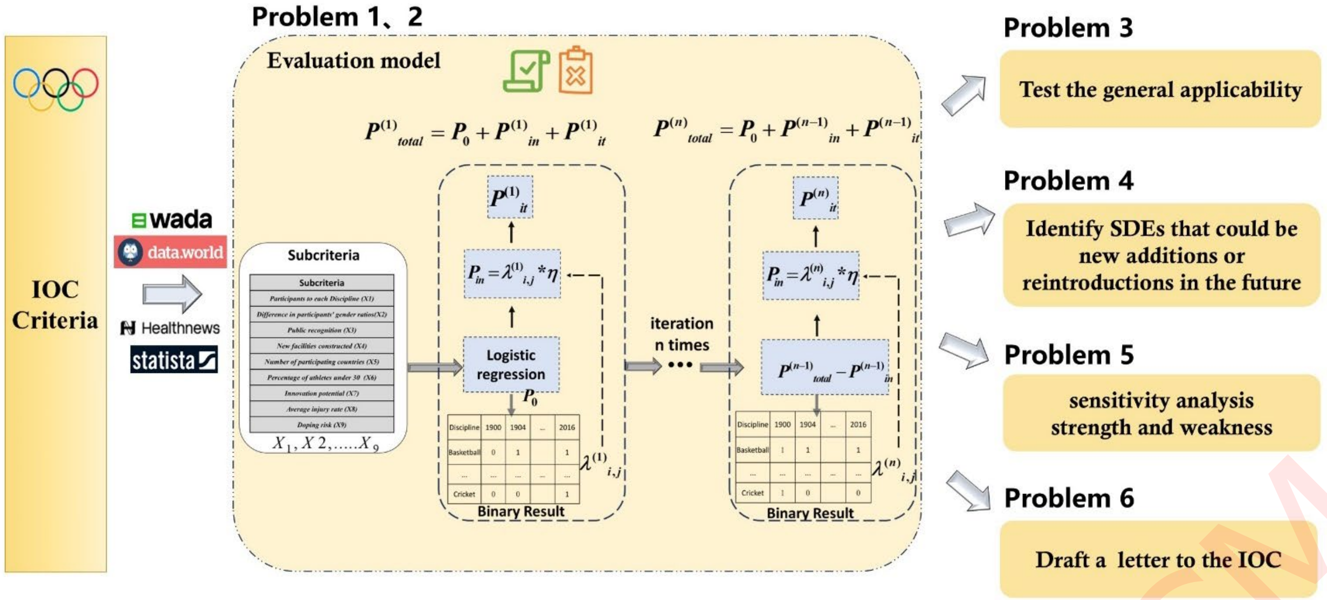

Figure 2 The workflow of our work

2. Assumptions and Justifications

Assumption 1: We assume that the nine subcriteria we have identified adequately capture the essential factors considered by the IOC.

Justification: While it is possible to expand the number of subcriteria, data availability poses a significant constraint [1]. Therefore, we have focused on a core set of factors that are both relevant and measurable. It is important to note that the performance of our model is contingent upon the quality and relevance of the input data. As more data becomes available, the model performance can be further improved.

Assumption 2: Due to data availability constraints, our analysis is primarily based on historical data up to 2016. We assume that the underlying trends and patterns observed in this period are indicative of future trends.

Justification: Given the time constraints of the contest, we were limited to collecting data up to 2016. By analyzing the historical data, we identified underlying patterns that can be extrapolated to make predictions for future Olympic Games. Recognizing the challenges associated with data collection, we opted to leverage a grey prediction model to extend the historical trends into the future. This approach is particularly suitable for scenarios where data is limited, and future trends are uncertain.

Assumption 3: We assume that no significant unforeseen events will occur during the forecast period that could substantially disrupt the Olympic Games or the inclusion of specific sports.

Justification: Historical data shows that while unforeseen events, such as extreme weather conditions, can lead to the cancellation of specific events (e.g., sailing and rowing in the 1896 Games), such occurrences are relatively rare within the broader context of the Olympic Games. Given the long history of the modern Olympics, the impact of such isolated incidents on the overall pattern of sport inclusion is considered negligible for our predictive model.

3. Evaluation criteria (Problem 1)

The IOC's Olympic Programme Commission has established evaluation criteria for SDEs, including Popularity and Accessibility, Gender Equity, Sustainability, Inclusivity, Relevance and Innovation, and Safety and Fair Play. To align with these six criteria, we have selected the following qualitative and quantitative subcriteria, as shown in Table 1.

Table 1 Established subcriteria influencing the inclusion of an SDE in the Olympics

IOC **criteria**	**Subcriteria**	**Unit**	**Data type**	**Range**
Popularity and Accessibility	Participants to each Discipline (X1)	persons	quantitative	[0,2508]
Gender Equity	Difference in participants' gender ratios (X2)	%	quantitative	[-100,100]
Sustainability	Public recognition (X3)	%	quantitative	[0,50.06]
	New facilities constructed (X4)	%	quantitative	[0,53.92]
Inclusivity	Number of participating countries (X5)		quantitative	[0,201]
Relevance and Innovation	Percentage of athletes under 30 (X6)	%	quantitative	[0,100]
	Innovation potential (X7)		qualitative	(low,middle, high)
Safety and Fair Play	Average injury rate (X8)	%	quantitative	[0,34]
	Doping risk (X9)	%	quantitative	[0,1.2]

For Popularity and Accessibility, we use the number of participants in each SDE as a quantitative metric. The IOC has previously categorized sports based on popularity, considering factors such as ticket sales, television viewership, and media coverage [2-3]. Figure 3 shows that popular sports, such as athletics and swimming, tend to have higher participation numbers. Therefore, the number of participants can serve as a proxy for popularity and accessibility.

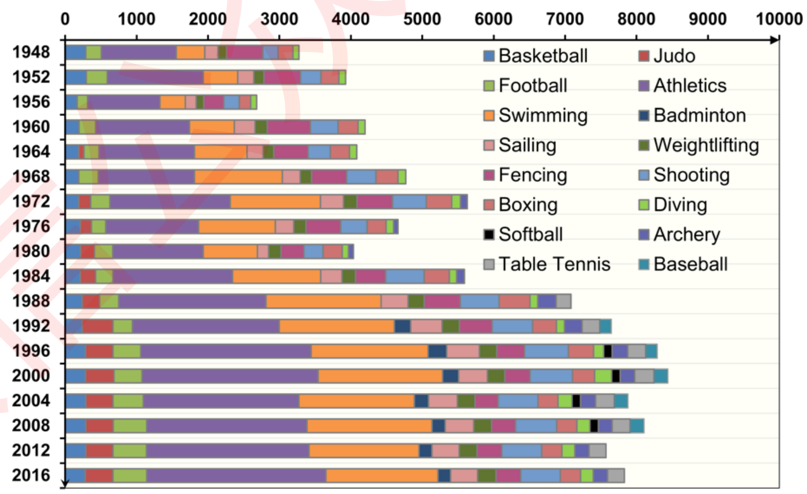

Figure 3 Temporal changes in the numbers of participants to the Olympics (data source as described in Section 4.1)

We selected the gender ratio difference within each SDE as an indicator of gender equality. In recent years, the Olympic movement has accelerated efforts to promote gender equality both on and off the field, thanks in part to progressive initiatives by the IOC. As Table 2 shows, the proportion of female athletes increased from 0% to 10.5% between 1896 and 1952. This trend continued to accelerate in subsequent decades. By the 2020 Tokyo Olympics, female participation reached a new record of 48%, and the 2024 Paris Olympics achieved complete gender parity [4]. As gender equality awareness grows, the gender ratio in different sports is converging, with the difference between male and female participation rates approaching zero. Therefore, to reflect the gender equality of different disciplines, we use the gender ratio difference.

Table 2 Female participants ratios in the 1896-2024 Olympic Games (data source as described in Section 4.1)

Olympic Games	Athens 1896	Paris 1900	St.Louis 1904	London 1908	Stockholm 1912	Antwerp 1920
Female ratio (%)	0	2.2	0.9	1.8	2	2.5
Olympic Games	Paris 1924	Amsterdam 1928	Los Angeles 1932	Berlin 1936	London 1948	Helsinki 1952
Female ratio (%)	4.4	9.6	9.5	8.4	9.5	10.5
Olympic Games	Melboume 1956	Rome 1960	Tokyo 1964	Mexioo City 1968	Munich 1972	Montreal 1976
Female ratio (%)	11.3	11.4	13.2	14.2	14.8	20.7
Olympic Games	Moscow 1980	Los Angeles 1984	Seoul 1988	Barcelona 1992	Atlanta 1996	Sydney 2000
Female ratio (%)	21.5	22.9	26.1	28.9	34	38.2
Olympic Games	Athens 2004	Beijing 2008	London 2012	Rio 2016	Tokyo2 020	Paris 2 024
Female ratio (%)	40.7	42.4	44.2	45	47.8	50

Sustainability in the context of the Olympics refers to the organizing committee's commitment to hosting the Games in an environmentally responsible, economically viable, and socially inclusive manner. The IOC emphasizes sustainability to ensure that the Olympics contribute to the UN's Sustainable Development Goals and address global environmental challenges [5]. Public recognition of the Olympics reflects the host's commitment to social responsibility, as governments that provide better social welfare and security during the Games tend to receive higher approval ratings. The Olympics can have a profound environmental impact on host cities. Constructing new venues and infrastructure often requires significant land use and can lead to habitat destruction, pollution, and increased carbon emissions. To mitigate these impacts, host cities should prioritize the use of existing facilities, invest in sustainable construction practices, and take measures to reduce resource consumption. Therefore, we selected public recognition of the event and the proportion of new facilities constructed as indicators of sustainability.

Inclusivity measures a sport's ability to attract a large number of participants from diverse geographical and cultural backgrounds (at least 75 countries across four continents practicing the sport). To assess the level of inclusivity, we use the number of participating countries. A higher number of participating countries indicates broader international recognition and greater cultural diversity. In addition to traditional sports like athletics and swimming, which have a large number of participating countries, sports like Judo, Karate, and Taekwondo represent different regional cultures (Figure 4a).

For the Relevance and Innovation criteria, we selected the percentage of athletes under 30 and the innovation potential of different SDEs. Attracting youth to the Olympic movement is one of the three main themes of Olympic Agenda 2020. Therefore, the proportion of young people reflects the relevance criterion. Figure 4b shows that the percentage of athletes under 30 in different SDEs has been increasing over time. We also conducted expert rating of the future innovation potential of different SDEs to reflect modern trends, such as the inclusion of breaking dancing in the 2024 Paris Olympics, which reflects the Olympic movement's focus on youth.

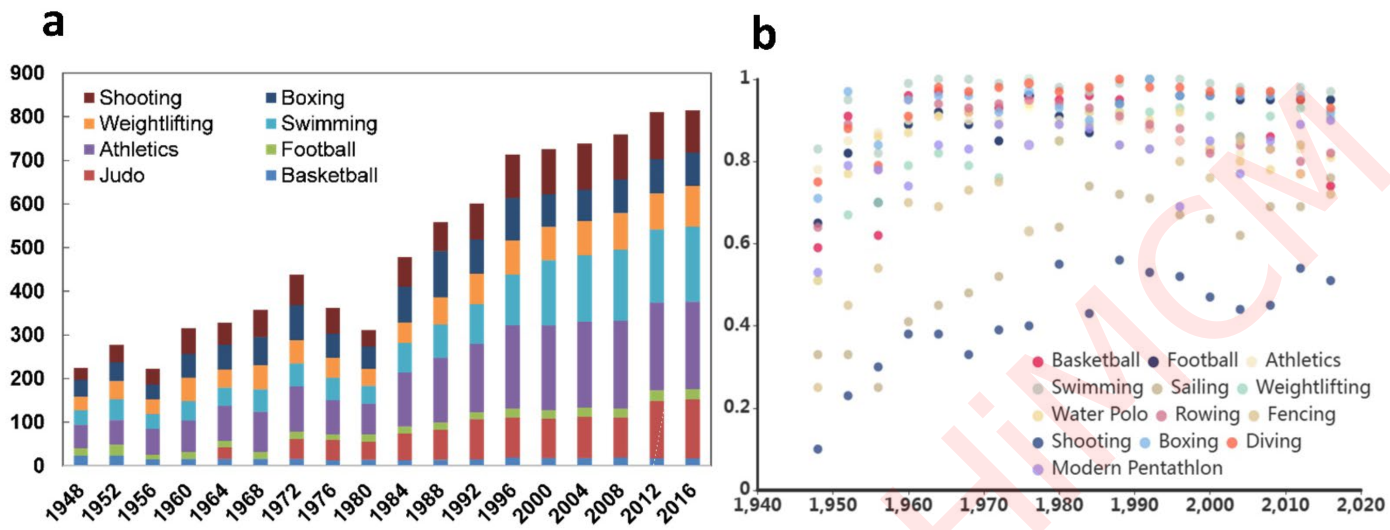

Figure 4 Variations in participating countries (a) and the proportion of athletes under 30 years old (b) across various SDEs (data source as described in Section 4.1)

"It takes more than crossing the line first to make a champion. A champion is more than a winner. A champion is someone who respects the rules, rejects doping and competes in the spirit of fair play." - Jacques Rogge, IOC President. Fair competition is essential in sporting. Fair play means not cheating through the use of drugs or doping. Therefore, for the fairness criterion, we selected the risk of doping in different SDEs. For safety, we focused on risk management in sports, aiming to prevent injuries, protect athletes' health, and take proactive measures to eliminate potential hazards in competitions. Therefore, we selected the injury risk of different SDEs as an indicator of safety. Figure 5 shows that sports like Cycling-BMX, football, and boxing have higher injury risks [6], while sports like shooting, diving, and Rhythmic Gymnastics have relatively lower injury risks.

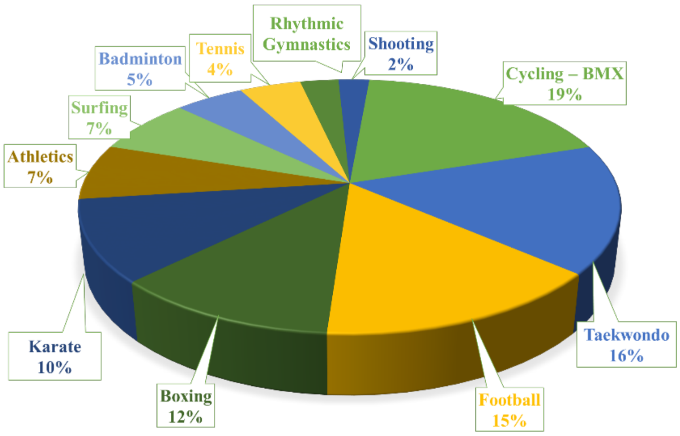

Figure 5 Aveage jury risks of selected SDEs (data source as described in Section 4.1)

4. Materials and Methodology

4.1 Data Collection and Preprocessing

To support our modelling, we collected data on nine subcriteria (Table 3) from various sources. Specifically, data on athlete demographics (gender, age, nationality, and discipline) for the period of 1900 to 2016 was sourced from Data World. This data allowed us to calculate variables such as the number of participants per discipline (X1), gender ratio difference (X2), the number of participating countries (X5), and the percentage of athletes under 30 (X6). Information on new venue construction (X3) and public support (X4) for each Olympic Games was obtained from the literature [1]. Injury risk data (X8) was collected from Healthnews, and doping rate (X9) was sourced from World Anti-Doping Agency (WADA).

Given the limited systematic research on the innovation potential of major sports (X7), we assigned scores based on expert opinion, considering factors such as technological integration like virtual reality. To predict potential sports for the 2032 Brisbane Olympics, we incorporated data from the Australian database, AUSPLAY, which provided information on injury risk (X8) and participant numbers (X1) for popular Australian sports.

Table 3 Data sources providing data for established subcriteria. Variables X1~X9 are referred to Table 1.

Database	Variables	Source
Data World	*X1, X2, X5, X6*	https://data.world/
Müller et al., 2021 [1]	*X3, X4*	https://doi.org/10.1038/s41893-021-00696-5
Healthnews	*X8*	https://healthnews.com/
WADA	*X9*	https://www.wada-ama.org/
AUSPLAY	*X1, X8*	https://www.clearinghouseforsport.gov.au/

To prepare the data for our model, we aggregated individual athlete data by discipline, calculating influential variables such as gender ratios, age demographics, and participation numbers. We also accounted for the evolving nature of the IOC's criteria, particularly in recent decades, by adjusting the data for earlier Olympic Games. This involved using time series extrapolation or averaging techniques to estimate missing or incomplete data. Ultimately, we created a dataset comprising 28 disciplines from 27 Olympic Games (1900 and 2016). These disciplines represent a diverse range. This includes long-standing disciplines that have been of the Olympic program since its inception (e.g., football, athletics, and swimming), sports that were popular in the early years but subsequently removed (e.g., polo and roque), as well as those that were reinstated after a period of absence (e.g., cricket and golf). Additionally, the dataset includes sports that were either non-competitive or newly introduced to the official program during the study period (1900-2016), such as triathlon, softball, and baseball. To ensure comparability, we normalized the data for each subcriterion.

4.2 Estimating Subcriteria Data for Gap Years and Future Olympics

The collected subcriteria data are limited to the period from 1900 to 2016. To address the data gap beyond 2016, we employed a grey prediction model, specifically the GM(1,1) model. This model is well-suited for systems with high uncertainty, making it suitable for forecasting the gap influential factor [7].

Step 1: The initial data sequence for each subcriterion was denoted as:

$$X^{(0)} = \left\{ x^{(0)}(1), x^{(0)}(2), \cdots, x^{(0)}(n) \right\} \tag{1}$$

Step 2: To ensure the suitability of the grey model, a ratio test was conducted on the original data. The ratio $\sigma(k)$ was calculated for each data point. If all ratios fell within the specified rage, the model could proceed.

$$\sigma(k) = \frac{x^{(0)}(k-1)}{x^{(0)}(k)}, \sigma(k) \in \left(e^{-\frac{2}{n+1}}, e^{\frac{2}{n+1}} \right) \tag{2}$$

Step 3: The accumulation generation operation was applied to the original data $x^{(0)}$ to weaken the random fluctuation and enhance the regularity. The resulting sequence was denoted as:

$$x_k^{(1)} = \sum_{i=1}^{k} x_i^{(0)}, k = 1, 2, \cdots, n \tag{3}$$

Step 4: The mean generation operation was applied to $X^{(1)}$ to generate a new sequence $Z^{(1)}$:

$$Z^{(1)} = \left\{ z^{(1)}(2), z^{(1)}(3), \cdots, z^{(1)}(n) \right\} \tag{4}$$

$$z^{(1)}(k) = \frac{1}{2}\left(x^{(1)}(k) + x^{(1)}(k-1) \right) \tag{5}$$

Step 5: The following GM(1,1) differential model was established:

$$x^{(0)}(k) + az^{(1)}(k) = b \tag{6}$$

where a and b are parameters to be estimated using the least square method.

Step 6: The estimated parameters were used to predict future values using the following equation:

$$\hat{x}^{(1)}(k) = \left[x^{(0)}(1) - \frac{b}{a} \right] e^{-a(k-1)} + \frac{b}{a}, \quad k = 1, 2.....n \tag{7}$$

4.3 Modeling SDE inclusion (Problem 2)

We treat the inclusion or exclusion of an SDE as a binary classification problem. To assess the probability of inclusion of certain discipline in the Olympics, we develop a logistic regression model that incorporates three key components (Eq. 8): a base probability (P_0) that evaluates the adherence to IOC's values based on nine quantitative and qualitative subcriteria we developed, an inertia component (P_{in}) accounting for the effect of the historical inclusion or exclusion of the SDE, reflecting a "momentum effect", and a bias-correction item (P_{it}) that adjusts the models predictions to best match historical data.

$$P_{total} = P_0 + P_{in} + P_{it} \tag{8}$$

When $P_{total} > 0.5$, the discipline is included; when $P_{total} < 0.5$, it is not included. Notably, we do not build individual models for each discipline but instead create a model able to estimate any discipline, which can be achieved by training the model using historical data across all disciplines.

The logistic regression model is express as:

$$\ln \frac{P_0}{1-P_0} = \boldsymbol{w}^T \boldsymbol{x} + b \tag{9}$$

where P_0 is the base probability of SDE being included, $\boldsymbol{x} = (X_1, X_2......X_9)$ is a vector of values fro the nine subcriteria variables, $\boldsymbol{w} = (W_1, W_2.....W_9)$ is a vector of weights for the nine subcriteria corresponding weights, and b is the bias term. The weights w and the bias b are estimated using maximum likelihood estimation based on historical data.

The inertia component (P_{in}) is calculated as the product of a weight $\lambda_{i,j}$ and a scaling parameter η. The weight $\lambda_{i,j}$ is essentially the conditional probability of the current state (inclusion or exclusion) given the previous state. This probability is estimated from historical data (the provided HiMCM_Olympic_Data.xlsx). If the previous state was exclusion, $\lambda_{i,j}$ is set to zero. The hyperparameter η scales the effect of the inertia. The value of η can be optimized to best fit the model results to the binarized historical records.

$$P_{in} = \lambda_{i,j} * \eta \tag{10}$$

where subscripts (i, j) denotes i-th discpline at the j-th Olympic Games.

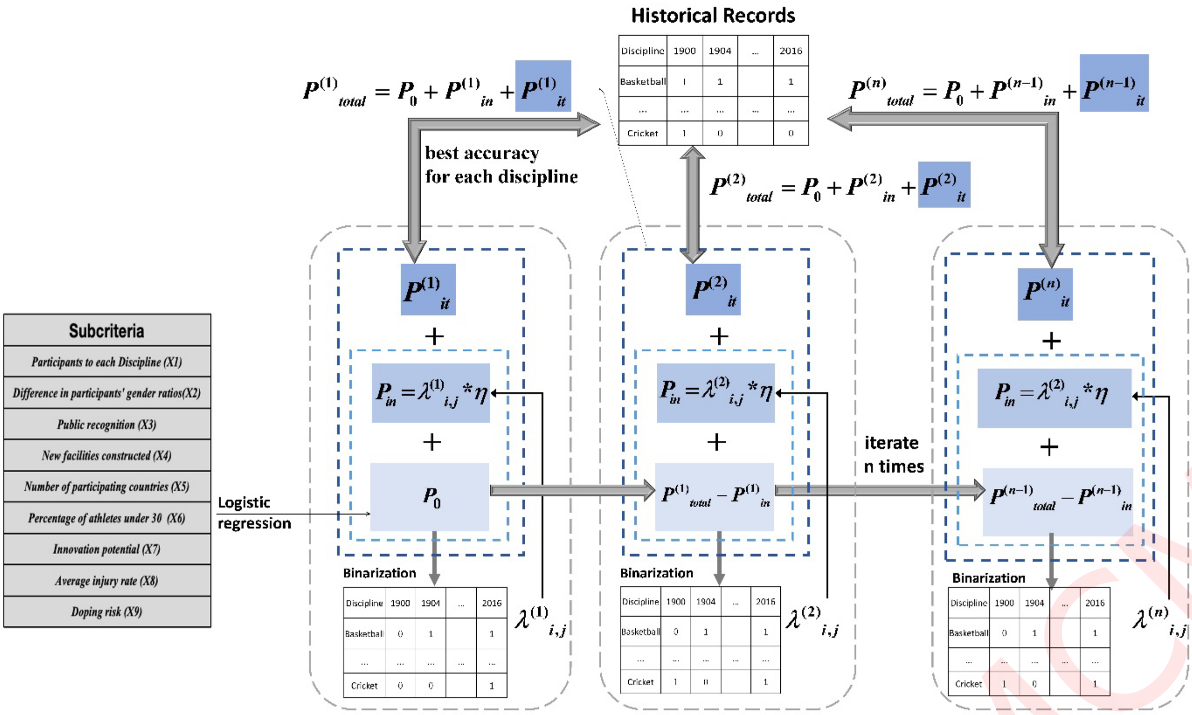

Figure 6 Diagram of the iterative modeling process

The model was implemented in Python and the iterative process is illustrated in Figure 6. The bias-correction component P_{it} is iteratively updated to improve the model's accuracy. In each iteration, the model's predictions are compared to the actual historical records, and P_{it} for each discipline is adjusted by a small increment (0.001). The direction of the adjustment (increasing or decreasing) depends on whether the model's prediction underestimates or overestimates the actual historical records.

Initially, the inertia weight $\lambda_{i,j}$ was estimated based on the classification results from the base probability P_0. In subsequent iterations, $\lambda_{i,j}^{(n)}$ was updated based on the classification results calculated from $P_{total}^{(n-1)} - P_{in}^{(n-1)}$. The iterative process continued until the model's accuracy converged to a threshold of 0.001 or a maximum of 1000 iterations was reached. To optimize the hyperparameter η, a grid search was conducted over the range of 0 to 1 with a step size of 0.1. For each value of η, the iterative process was performed, and the optimal value was selected based on the model's accuracy.

4.4 Model Testing (Problem 3)

To demonstrate the generality of the proposed model, we evaluate it from three aspects: (1) cross-validation, (2) validation for newly included/reintroduced disciplines, and (3) validation for long-standing disciplines. Cross-validation helps reduce the evaluation bias introduced by random data partitioning, thereby providing a more stable estimate of the model's performance. Considering the sample size (28 disciplines × 27 Olympic Games), we conduct 5-fold cross-validation for the logistic regression. Specifically, the dataset from 1900 to 2016 is randomly divided into 5 parts. Among them, 4 parts are used to train the model, and the remaining 1 part is used for validation. The train-validation mode is repeated for 5 times to assure every part can be used as validation part. Then the averaged accuracy can be reliably represented the model performance.

For the validation of newly included/reintroduced disciplines in 2020, 2024, and 2028 Olympic Games, we use the model developed to make predictions, where the corresponding subcriteria for 2024-2028 are generated using the grey forecasting model as described in Section 5.2. To assess the model's performance on long-standing disciplines, we arbitrarily select three common disciplines (Swimming, Athletics, Archery) from the historical records between 19884 and 1992. To ensure a rigorous evaluation, we intentionally excluded these testing disciplines from the training dataset and trained a separate model for each.

During the model performance evaluation, we primarily use the Overall Accuracy metric (i.e., the ratio of correct predictions to total predictions), which is commonly used in binary classification problems. Additionally, we use Precision, Recall, and F1-Score to assess the cross-validation results.

4.5 Influential Factors Analysis (Problem 5)

To quantify the impact of subcriteria uncertainty on model accuracy, we applied ±2.5%, ±5.0%, ±7.5%, and ±10.0% perturbations to the model developed in Section 5.3 and examined the relative change in model accuracy after these perturbations. Additionally, during the model building process, we performed uniform sampling within the range of the hyperparameter η to obtain the best accuracy for each η, of which the trajectory can be used to measure its influence.

4.6 Prediction for the 2032 Olympics and Beyond (Problem 4)

When considering potential SDEs for the 2032 Brisbane Olympics, it's important to account for the host country's sporting culture and global trends. Recent examples like the inclusion of Karate in the 2020 Tokyo Olympics and Breaking in the 2024 Paris Olympics highlight the influence of host nations. Based on data from AUSPLAY, two potential SDEs that could be considered for the 2032 Brisbane Olympics stand out from the 9 most popular disciplines in Australia (Table 4).

Australian Football is a highly acclaimed national sport in Australia with audiences reaching 6.75 million, of which over 1/10 younger than 30 years old. Its sustainability efforts are notable, as shared facilities with cricket fields help reduce carbon emissions. Recently, the sport has been rapidly expanded to over 20 countries [8].

Netball attracts more than 1.2 million players across various age groups and genders, establishing itself as Australia's largest team sport. Its league can draw 0.12 million spectators per match. While traditionally a women's sport, netball is increasingly gaining popularity among men. It is also a globally popular sport, played in over 80 countries with more than 20 million participants [9].

We also include Zwift cycling as a global trend, which was mentioned in the contest problem. Zwift cycling merges traditional cycling with virtual technology, boasting over 3 million users or participants across 190 countries since its launch in 2015. The platform appeals to a young, offering accessible, low-cost, and safe participation.

Additionally, we considered the potential reintroduction of sports that have previously been part of the Olympic program. Based on recent trends, such as the return of baseball and softball to the 2028 Los Angeles Olympics, sports like polo, croquet, roque, and Basque pelota could be potential candidates for future inclusion. As a results, we identified a total of 7 candidate disciplines to be assessed for inclusion in future Olympics 2032 and beyond.

Table 4 Top 9 most popular sports in Australia [10]

Rank	**Sport**	**Popularity**	**Included in the Olympics**
1	Australian Football	75.7 Million Fans	No
2	Cricket	14.0 Million Fans	Yes
3	Tennis	12.5 Million Fans	Yes
4	Netball	6.7 Million Fans	No
5	Swimming	6.0 Million Participants	Yes
6	Cycling	3.4 Million Participants	Yes
7	Football	2.0 Million Participants	Yes
8	Basketball	1.3 Million Participants	Yes
9	Rugby Union	1.3 Million Fans	Yes

5. Results and Analyses

5.1 The Evaluation Model (Problem 2)

The initial component of our model was estimated using a logistic regression. Table 5 presents the fitted weights of the nine subcriteria variables. A variance inflation factor (VIF) below 3 for all variables indicates the absence of multicollinearity, satisfying a fundamental assumption of logistic regression. Furthermore, all coefficients were statistically significant ($p < 0.05$), suggesting that each variable contributed meaningfully to the model.

Specifically, while new facility construction (X4) and doping risk (X9) had negative impacts, other variables had positive effects. The most influential variable was the number of participants per discipline (X1), followed by the difference in participants' gender ratios (X2) and the number of participating countries (X7). These findings align with the IOC's selection criteria, as increasing X4 suggests unsustainability, while X9 indicates a higher risk of unfair competition. Conversely, higher values of X1, X2, and X7 align with the IOC's preference for globally impactful sports.

Table 5 The fitted parameters for the first component (P_0) in the established model which is a logit function

Subcriteria	***w***	**p-value**	**VIF**
Participants to each Discipline (X1)	1.33	0.006	1.065
Difference in participants' gender ratios (X2)	0.87	0.003	1.145
Public recognition (X3)	0.24	0.008	1.546
New facilities constructed (X4)	-0.85	0.002	1.782
Number of participating countries (X5)	0.81	0.006	1.264
Percentage of athletes under 30 (X6)	0.72	0.045	1.105
Innovation potential (X7)	0.10	0.032	1.245
Average injury rate (X8)	0.32	0.039	1.463

Doping risk (X9)	-1.88	0.024	1.364
Bias	0.10	0.029	

Iterative optimization significantly improved model accuracy from 0.833 for the first estimation to 0.952 (Figure 7), reducing misclassifications from 125 to 36. Of these errors, 17 were false negatives, suggesting that the model was more likely to underestimate the inclusion of new disciplines. The optimal hyperparameter value, $\eta = 0.47$, balanced the influence of historical data and new information. The final values of P_{it} (Table 6) indicate that the estimates initially overestimated the likelihood of inclusion, but the bias correction term effectively mitigated this bias.

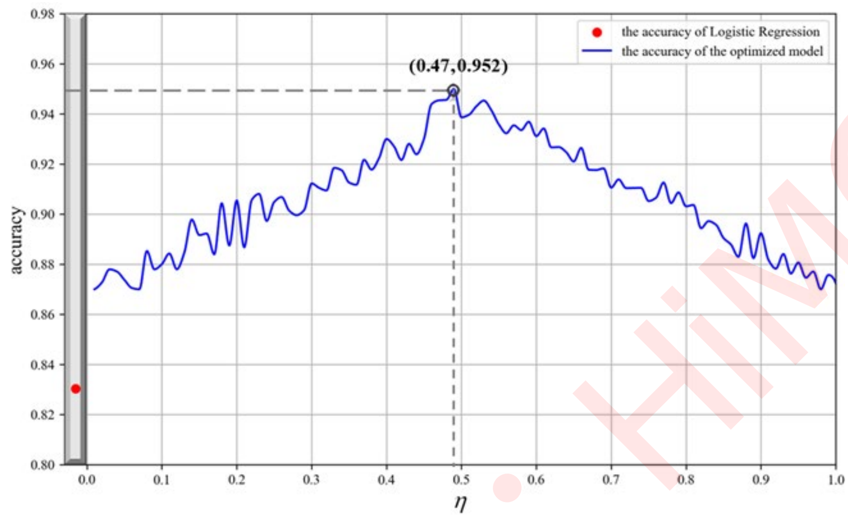

Figure 7 Model's accuracies varying with the value of parameter η which scales the effect of inertia

Table 6 The final values of P_{it} associated with each discipline

Discipline	P_{it}	Discipline	P_{it}	Discipline	P_{it}	Discipline	P_{it}
Basketball	-0.006	Shooting	-0.003	Triathlon	-0.068	Weightlifting	0.000
Judo	-0.007	Boxing	-0.003	Polo	-0.069	Water Polo	-0.004
Football	-0.002	Taekwondo	-0.069	Cricket	-0.069	Softball	-0.069
Athletics	-0.422	Diving	-0.162	Croquet	-0.069	Archery	-0.017
Swimming	-0.422	Tennis	-0.011	Roque	-0.068	Fencing	0.000
Badminton	-0.011	Modern Pentathlon	-0.332	Basque Pelota	-0.069	Table Tennis	-0.007
Sailing	0.000	Golf	-0.069	Rowing	-0.02	Baseball	-0.009

5.2 Performance in Predicting Historical Patterns (Problem 3)

In the case of recently adjusted inclusion of disciplines, our model's predictions perfectly aligned with actual outcomes (Table 7). For newly introduced Baseball and Cricket, the predicted probabilities of inclusion were 0.592 and 0.549, respectively, exceeding the 0.5 threshold. Conversely, for the excluded Softball, the predicted probability was 0.365.

Regarding long-standing cases, our model consistently predicted inclusion probabilities above 0.5 (Table 8). Interestingly, we observed a notable difference in predicted probabilities between recently added and long-standing disciplines. Newly added disciplines had lower probabilities (0.5-0.6) compared to long-standing disciplines (≥0.7), primarily due to lower scores in Popularity, Accessibility, and Inclusivity.

These findings strongly support the high predictive accuracy of our proposed model, which effectively captures both long-standing and recently changed SDEs. It's worth noting that for recent changes, we relied on a grey forecasting model to estimate missing subcriteria. This further demonstrates our model's robustness to data uncertainties.

Table 7 Validation for introduced or removed disciplines in recent Olympic Games

subcriteria	**Baseball**	**Cricket**	**Softball**
	2020	2028	2024
X1	191	53	120
X2(%)	100	100	0
X3(%)	37.58	42.11	44.90
X4(%)	14.47	15.68	15.81
X5	8	7	8
X6(%)	80	45	75
X7	low	low	low
X8(%)	11.00	8.48	13.00
X9(%)	0.10	0.07	0.04
True label	1	1	0
Predicted	0.592	0.549	0.365

Table 8 Validation for long-standing disciplines

	Discipline	**1984**	**1988**	**1992**
True Label	Swimming	1	1	1
Predicted	Swimming	0.789	0.839	0.768
True Label	Athletics	1	1	1
Predicted	Athletics	0.852	0.869	0.874
True Label	Archery	1	1	1
Predicted	Archery	0.753	0.761	0.796

5.3 Prediction for the 2032 Brisbane Olympics (Problem 4):

Table 9 presents the predicted probabilities of inclusion for the seven candidate disciplines in the 2032 Brisbane Olympics, as determined by our model. The results indicate a strong likelihood of Australian Football being included, with a predicted probability of 0.857. This aligns with the sport's growing global popularity and its deep roots in Australian culture. Zwift Cycling and Netball also have relatively high probabilities, reflecting their increasing participation rates and potential to attract a wide audience.

In contrast, traditional sports like Polo, Croquet, Roque, and Basque Pelota have lower predicted probabilities, suggesting that they may face challenges in being reintroduced to the Olympic program. This could be due to factors such as declining popularity, limited global reach, or specialized infrastructure requirements.

These findings highlight the potential of our model to identify promising candidates for future Olympic inclusion based on a range of factors. However, it is important to note that the final decision on which sports to include in the Olympics is ultimately made by the International Olympic Committee, and may be influenced by additional political, social, and economic considerations.

Table 9 Prediction of inclusion of 7 potential disciplines in the 2032 Brisbane Olympics

Discipline	Australian Football	Netball	Zwift Cycling	Polo	Croquet	Roque	Basque Pelota
X1	220	140	120	21	19	4	2
X2 (%)	60	-74	20	100	37	100	100
X3 (%)	40	30	50	0	0	0	0
X4 (%)	20	10	5	0	0	0	0
X5	20	80	150	5	2	2	2
X6 (%)	76	76	55	24	37	11	100
X7	middle	middle	high	low	low	low	low
X8 (%)	6	5	2	3.36	0	0	0
X9 (%)	0.047	0.047	0.047	0.047	0.047	0.047	0.047
Predicted	0.757	0.536	0.710	0.327	0.288	0.274	0.255
Rank	1	3	2	4	5	6	7

5.4 Prediction for 2036 and Beyond Olympics (Problem 4)

Using the grey forecasting method, we projected future values of the nine subcriteria for the top three disciplines in 2032 (Australian Football, Zwift Cycling, and Netball) up to 2040. These projected values were then fed into our model to predict future inclusion probabilities. As shown in Figure 8, Zwift Cycling exhibits a consistently increasing trend from 2032 onwards, while Australian Football and Netball display a decreasing trend followed by a slight increase. Notably, Zwift Cycling demonstrates the highest predicted inclusion probability, exceeding 0.85 from 2036 onward. Based on these results, we can infer that Zwift Cycling is the most likely of the three to be included as an Olympic event after 2036.

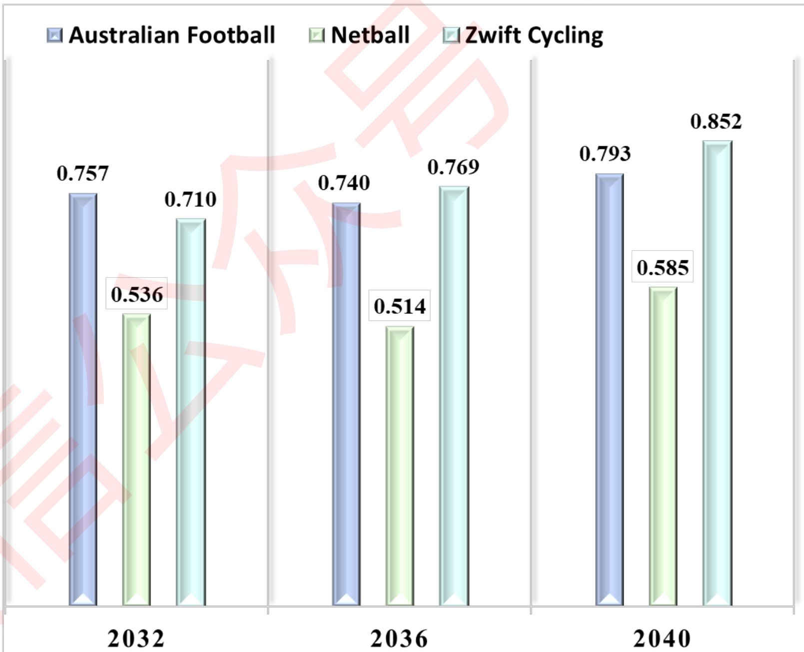

Figure 8 Modelled probability of including Australian Football, Netball and Zwift cycling in 2032, 2036 and 2040 Olympic Games

6. Sensitivity Analysis

6.1 Influential Factors for Inclusion of an Olympic Sport (Problem 5)

We conducted a sensitivity analysis on the nine evaluation criteria listed in Table 1 by systematically varying each criterion while holding the others constant. Figure 9 illustrates the changes in model accuracy when perturbing each subcriterion by ±2.5%, ±5.0%, ±7.5%, and ±10.0%. Our findings indicate that when the perturbation exceeds 7.5%, the model's accuracy is primarily affected by the number of participants per discipline and the proportion of participants under 30 years old. As the perturbation increases to more than 5%, the model's accuracy becomes more sensitive to the number of participating countries and the gender ratio difference, leading to an approximate 0.02 decrease in accuracy. For perturbations smaller than 5%, the model's predictions are less sensitive to changes in a discipline's innovative potential, public recognition, new facility construction, and doping risk. Overall, the model's accuracy remains relatively stable across various perturbation levels, demonstrating its robustness and ability to provide reliable assessments of different disciplines.

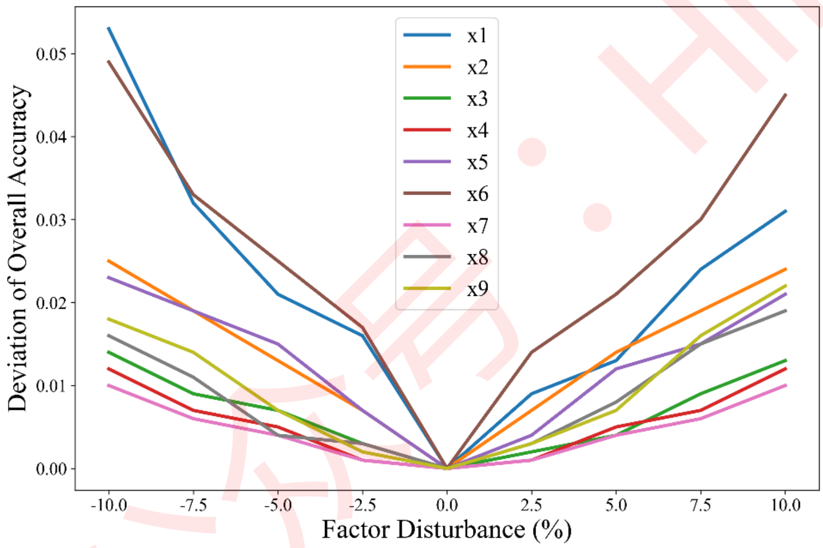

Figure 9 Sensitivity analysis on the influential factors as inputs to the model. Participants to each Discipline (X1), Difference in participants' gender ratios (X2) , Public recognition (X3) , New facilities constructed (X4), Number of participating countries (X5), Percentage of athletes under 30 (X6), Innovation potential (X7), Average injury rate (X8), Doping risk (X9).

6.2 Sensitivity on model parameters

We also conducted a sensitivity analysis on the inertia scaling parameter $\boldsymbol{\eta}$, which varies between 0 and 1, to evaluate its impact on model accuracy. As shown in Figure 7, the optimal model accuracy is achieved when $\eta = 0.47$. For values between 0 and 0.47, the model accuracy exhibits a fluctuating increase as η increases. Conversely, for values between 0.47 and 1, the accuracy fluctuates but with a decreasing trend. Overall, the model's sensitivity to η is relatively low, with the maximum decrease in accuracy being only 0.08 within the specified range. This indicates that the model is quite robust. The model's robustness can be attributed to the bias-correction term incorporated into our model, which effectively compensates for the potential decrease in accuracy caused by variations in η.

7. Strength and Limitations

Key Strengths:

1. Our model demonstrates a high level of accuracy in predicting the inclusion or exclusion of SDEs. It successfully captures historical trends and patterns.

2. The model's performance is relatively insensitive to noise or uncertainty in the input data, ensuring reliable predictions even under varying conditions.

3. The model incorporates a comprehensive set of criteria, including popularity, gender equity, sustainability, inclusivity, relevance, innovation, safety, and fair play, to provide a holistic assessment of SDEs.

4. The model leverages historical data to learn patterns and make informed predictions, reducing reliance on subjective expert judgments. This data-driven approach ensures objectivity and consistency in the evaluation process.

5. The model can be easily adapted to incorporate new data and emerging trends, making it a valuable tool for future decision-making. Additionally, the model's modular structure allows for customization and refinement as needed.

Limitations:

1. The model's accuracy is dependent on the quality and relevance of the input data. The nine subcriteria used in this study, while comprehensive, may not fully capture all the nuances of the IOC's decision-making process. Additional factors, such as geopolitical considerations or specific host city requirements, could influence the inclusion of SDEs.

2. The analysis was based on 28 disciplines, which may not be fully representative of the entire universe of potential Olympic SDEs. A larger size would allow for more robust modeling and potentially improved predictive accuracy.

3. The IOC's criteria for SDE selection have evolved over time, with increasing emphasis on factors like sustainability and gender equality in recent years. Our model may not fully account for these historical changes, potentially affecting its predictive accuracy for future Olympics.

8. Conclusions

We developed a probability-based model to predict the inclusion of sports, disciplines, and events (SDEs) in future Olympic Games, aligning with the International Olympic Committee's (IOC) criteria. The model integrates a base probability (P0) assessing IOC value adherence, an inertia component (Pin) accounting for past inclusion, and a bias-correction item (Pit) to optimize historical data fit. Sensitivity analysis highlighted the importance of Popularity and Relevance and Innovation criteria, with predictions for the 2032 Brisbane Olympics suggesting Australian football, Zwift cycling, and netball as leading candidates.

Our model demonstrates high predictive accuracy and provides a data-driven alternative to traditional, subjective methods. By learning from historical trends, the model offers objective and flexible predictions while being adaptable to more granular levels with sufficient data. Challenges in data collection remain, but future efforts could further refine the model's scope and enhance its reliability for evaluating Olympic SDE inclusions.

References

[1] Müller, M., Wolfe, S.D., Gaffney, C. *et al.* An evaluation of the sustainability of the Olympic Games. *Nat Sustain* **4**, 340–348 (2021). https://doi.org/10.1038/s41893-021-00696-5

[2] "Athletics to share limelight as one of top Olympic sports". The Queensland Times. 31 May 2013. Retrieved 18 July 2013.

[3] Murray, Callum (29 May 2013). "Winners Include Gymnastics, Swimming - and Wrestling - as IOC Announces New Funding Distribution Groupings". ASOIF.com. St Petersburg: Association of Summer Olympic International Federations. Archived from the original on 14 July 2014. Retrieved 18 July 2013.

[4] Gender equality through time.https://olympics.com/ioc/gender-equality/gender-equality-through-time

[5] https://library.olympics.com/Default/doc/SYRACUSE/169337/olympic-review-official-publication-of-the-olympic-movement-vol-63-april-may-june-2007?_ga=2.237747717.461168666.1681109728-862628506.1677753585

[6] Most Dangerous and Safest Olympic Sports by Injury Rates (8 May 2024). Edvardas Garbenis. https://healthnews.com/reports/most-dangerous-and-safest-olympic-sports-by-injury-rates/

[7] Ding S., Dang Y. G., Xu N., Wang J. J., and Xu Z. D., A novel grey model based on the trends of driving factors and its application, *The Journal of Grey System*. (2018) **30**, no. 3, 105–126.

[8] Most Dangerous and Safest Olympic Sports by Injury Rates (8 May 2024). Edvardas Garbenis. https://healthnews.com/reports/most-dangerous-and-safest-olympic-sports-by-injury-rates/

[9] Thompson, Shona M. (December 2002). "Women and sport in New Zealand". In Pfister, Gertrud; Hartmann-Tews, Ilse (eds.). *Sport and Women: Social Issues in International Perspective*. International Society for Comparative Physical Education & Sport. New York: Routledge. ISBN 0-415-24628-8. OCLC 50204306

[10] https://www.clearinghouseforsport.gov.au/research/ausplay/results-2015-2023

Appendices

A The established model in Python

```python
import pandas as pd
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

'''calculate P_0'''
model = LogisticRegression()
model.fit(X, y)
y_sigmoid = model.predict_proba(X)[:, 0]
y_sigmoid2 = model.predict(X)

'''calculate lambda'''
def lambda_func(P0_bina):
    years = P0_bina.columns
    num01, num11 = 0, 0
    for i in range(len(years) - 1):
        year1 = years[i]
        year2 = years[i + 1]
        P0_bina = P0_bina.reset_index(drop=True)
        changes = P0_bina[year2] - P0_bina[year1]
        num11 += ((P0_bina[year1] == 1) & (P0_bina[year2] == 1)).sum()
    lambda_01 = num01 / (num01 + num11)
    lambda_11 = num11 / (num01 + num11)
    return lambda_01, lambda_11
def tell_lambda(x, lambda_01, lambda_11, eta):
    if x == 0:
        return lambda_01 * eta
    elif x == 1:
        return lambda_11 * eta
    else:
        raise ValueError("Invalid value in the data")

'''calculate P_inertia'''
def Pin_func(P0_bina, eta):
    lambda_01, lambda_11 = lambda_func(P0_bina)
    bina_shifted = P0_bina.shift(periods=1, axis=1)
    bina_shift2 = bina_shifted.iloc[:, 1:]
    Pin = bina_shift2.applymap(lambda x: tell_lambda(x, lambda_01, lambda_11, eta))
    return Pin, lambda_11

'''calculate P_iteration'''
def find_optimal_Pit(real_row, simulated_row, Pit_range):
    best_accuracy = 0
    best_constant = None
    for constant in Pit_range:
        constant = np.round(constant, decimals = 2)
        adjusted_simulated_row = (simulated_row + constant).dropna()
        if (0 < adjusted_simulated_row).all() and (adjusted_simulated_row < 1).all():
            bina_simulated_row = Binarize(adjusted_simulated_row, threshold)
            accuracy = accuracy_score(real_row, bina_simulated_row)
            if accuracy > best_accuracy:
                best_accuracy = accuracy
                best_constant = constant
    return best_constant, best_accuracy
def Pit_func(real_df, simulated_df):
    Pit_results = []
    for index, real_row in real_df.iterrows():
        simulated_row = simulated_df.iloc[index]
        best_constant, best_accuracy = find_optimal_Pit(real_row, simulated_row, Pit_range)
        Pit_results.append((index, best_constant, best_accuracy))
    return Pit_results
```

```python
'''calculate P_total'''
def Ptot_func(pP0_prob, eta):
    pP0_bina = Binarize(pP0_prob, threshold)
    pP0_bina.insert(loc = 0, column = "1900", value = bina_1900)
    pP0_bina1 = pP0_bina
    Pin_df, lambda_11 = Pin_func(pP0_bina1, eta)
    sSimulated_df = pP0_prob + Pin_df
    Pit_result = Pit_func(real_bina, sSimulated_df)
    Pit_result = pd.DataFrame(Pit_result,
                              columns=['index', 'best_constant', 'best_accuracy'])
    Pit_array = Pit_result['best_constant'].fillna(0).values
    Ptot = sSimulated_df + Pit_array[:, None]
    return Ptot, Pit_array, lambda_11

'''calculate model overall accuracy'''
def interation_Accuarcy(Ptot, real_bina):
    Ptot_bina = Binarize(Ptot, threshold)
    if real_bina.shape != Ptot_bina.shape:
        raise ValueError("The two dataframes must have the same shape.")
    true_values = real_bina.values.flatten()
    pred_values = Ptot_bina.values.flatten()
    overall_accuracy = accuracy_score(true_values, pred_values, average='binary')
    return overall_accuracy

'''Iteration Process'''
threshold = 0.5
Pit_range = np.arange(-0.005, 0.500 + 0.001, 0.001)
eta_range = np.arange(0.000, 1.000 + 0.001, 0.001)
pPit = np.zeros(28, dtype=np.float64)
a_rows = []
for a in eta_range:
    itera_results = []
    for i in range(1000):
        Ptot, Pit, lambda_11 = Ptot_func(P0_prob, a)
        P0_prob += Pit[:, None]
        pPit += Pit
        itera_results.append({'accuracy_result':interation_accuracy,
                              'eta':a, 'lambda_11':lambda_11, 'Pit_rows':pPit})
    max_accuracy = -float('inf')
    best_result = None
    for result in itera_results:
        if result['accuracy_result'] > max_accuracy:
            max_accuracy = result['accuracy_result']
            best_result = result

    a_rows.append(best_result)
    print(best_result)
```

A Letter to IOC

Dear Members of the IOC:

On behalf of HiMCM Olympic Consultants (HOC), we are honored to have been entrusted with the task of evaluating potential Sports, Disciplines, or Events (SDEs) for the 2032 Summer Olympics. Our team developed a sophisticated model to assess SDEs based on the IOC's criteria.

Our model analyzes various factors such as popularity, gender equality, sustainability, and global reach to predict the likelihood of an SDE's inclusion. By considering historical data and future trends, our model provides valuable insights into the potential impact of different SDEs on the Olympic Games.

Based on our analysis, we recommend the following SDEs for consideration in the 2032 Brisbane Olympics:

- Australian Football: A popular sport in Australia with a growing global following.
- Zwift Cycling: A virtual cycling platform that offers accessible and inclusive opportunities for athletes.
- Netball: A global sport with a strong female following and a growing male participation rate.

We believe that these SDEs align with the IOC's vision of promoting sportsmanship, excellence, and friendship. They have the potential to attract new audiences, enhance the diversity of the Olympic program, and contribute to the long-term success of the Olympic Games.

While traditional sports like polo, croquet, roque, and Basque pelota have historical significance, our model suggests that they may not be the best fit for future Olympics. This is due to factors such as declining popularity, limited global reach, and specialized infrastructure requirements.

Our model offers a data-driven approach to evaluating SDEs, providing a more objective and evidence-based decision-making process. While data limitations and unforeseen circumstances may influence future outcomes, our model provides a solid foundation for informed decision-making.

We are committed to further refining our model and providing ongoing support to the IOC. Please do not hesitate to contact us if you have any questions or require additional information.

Thank you for your time and consideration.

Yours sincerely,
The HiMCM team

AI Use Report

Google's Gemini language model was used solely to revise the language of the sections 1.1, 3 and 8, as well as the letter to IOC, building upon our draft. No generative AI tools were employed in the creation of the core content, methodology or programming.